

Business Problem Statement

A UK-based online retailer specializing in all-occasion gift-ware recorded 541,910 transactions between December 2010 and December 2011, across 4,372 identified customers, over 4,000 products, and sales to 38 countries. Despite this volume of data, the business has no structured view of which products and markets actually drive revenue, which customers are most valuable, or which are quietly disengaging. A meaningful share of transactions also come from unidentified guest/wholesale checkouts and order cancellations — both of which represent real, measurable revenue impact the business hasn't quantified.

You are tasked with analyzing the company's transaction dataset to answer the following overarching business question:

"How can the company use its transaction data to identify its strongest revenue drivers, understand customer value and loyalty, and reduce losses from cancellations and customer disengagement?"

Deliverables

1. **Data Preparation & Modeling (Python):** Clean the raw transaction data — recover missing product descriptions via lookup, remove duplicate and non-product (fee/postage) rows, and filter invalid prices — while deliberately preserving missing Customer IDs and cancellations as meaningful business signals rather than deleting them.
2. **Data Analysis (SQL):** Load the cleaned data into PostgreSQL and write queries to answer 10 business questions covering revenue by country, customer loyalty segmentation, cancellation patterns, and seasonal trends. Build reusable SQL views to feed the dashboard layer directly.
3. **Visualization & Insights (Power BI):** Build an interactive dashboard with KPI cards (revenue, customer count, average order value, cancellation rate), a customer segment breakdown, top revenue countries and products, and a monthly revenue trend, enabling stakeholders to explore patterns without writing SQL themselves.
4. **Report and Presentation:** Write a clear project report summarizing key findings (e.g., revenue concentration by country, the cost of cancellations, customer loyalty distribution) and business recommendations. Prepare a short presentation communicating the insights and recommendations.
5. **GitHub Repository:** Include all Python cleaning scripts, SQL queries and views, the Power BI dashboard file (or screenshots), and a README documenting the problem, approach, and findings in a well-structured repository.